

2 Description

The STM32C542xx devices are general purpose microcontrollers family (STM32C5 series) based on the high-performance Arm® Cortex®-M33 32-bit RISC core. They operate at a frequency of up to 144 MHz.

The Cortex®-M33 core features a single-precision floating-point unit (FPU) that supports all the Arm® single-precision data-processing instructions and all the data types.

The Cortex®-M33 core also implements a full set of digital signal processing (DSP) instructions and a memory protection unit (MPU) that enhances the application security.

The devices embed high-speed memories (up to 256-Kbyte flash memory and 64-Kbyte SRAM), and an extensive range of enhanced I/Os, peripherals connected to three APB buses, three AHB buses, and a 32-bit multi-AHB bus matrix.

The devices feature several protection mechanisms for embedded flash memory and SRAM: readout protection, write protection, and hide protection areas.

The devices embed several peripherals reinforcing security:

- HASH hardware accelerator
- True random number generator
- AES coprocessor

The devices offer one 12-bit ADC, two DAC channels, two comparators, one operational amplifier, a low-power RTC, one 32-bit general-purpose timer, two 16-bit PWM timers dedicated to motor control, two 16-bit general-purpose timers, two 16-bit basic timers, and one 16-bit low-power timer.

The devices also feature standard and advanced communication interfaces such as:

- One I²C
- One I³C
- Two SPIs with muxed full-duplex I2S
- Two USARTs, two UARTs, and one low-power UART
- Two FDCANs
- One USB full-speed

The devices operate in the -40 to +105 °C (+140 °C junction) temperature ranges from a 2.7 to 3.6 V power supply.

A comprehensive set of power-saving modes allows the design of low-power applications.

The devices offer multiple packages from 20 to 64 pins.

See [Table 1](#) for the list of peripherals available for each part number.

Table 1. Device features and peripheral counts

Peripherals		STM32C542FCP	STM32C542FCU	STM32C542ECU	STM32C542KCT	STM32C542KCU	STM32C542CCT STM32C542CCU	STM32C542RCT
Number of pins		20	20	24	32	32	48	64
Flash memory (Kbytes)		256						
SRAM (Kbytes)		64 (including 32 with ECC)						
ICACHE (Kbytes)		8						
Data flash memory (Kbytes)		64						
One-time-programmable (Kbytes)		4.5						
Timers	General purpose	1 (32-bit) and 2 (16-bit)						
	Advanced-control	2						
	Basic	2						